

Ayoub GHRISS

Efficient ML, Sparsity & Quantization, GPU Systems

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in ayghri

Scholar

MA, FR

PUBLICATIONS

Preprint	S³: Structured Sparsity Specification Ghriss, A. <i>Sparsity algebra unifying hardware-aligned patterns.</i>
Preprint	Kernelized Linear Attention: Breaking the Capacity Wall with Symmetric Cones Ghriss, A., Chakraborty, S. <i>Provable bounds on the capacity of linear attentions. Under review.</i>
AISTATS '26	Beyond Spectral Clustering: Probabilistic Cuts for Differentiable Graph Partitioning Ghriss, A. CU Boulder CS Dept. Outstanding Paper Award
ICML-W '25	Crowdsourced Information Authentication: A Graph-based Model from the Science of Hadith Ghriss, A. <i>MusML Workshop, ICML 2025 – oral.</i>
AISTATS '25	Deep Clustering via Probabilistic Ratio-Cut Optimization Ghriss, A., Monteleoni, C. CU Boulder CS Dept. Outstanding Paper Award
ASCE '24	Predicting Serious Injury and Fatality Exposure using Machine Learning in Construction Projects Oguz Erkal, E.D., Hallowell, M.R., Ghriss, A., Bhandari, S.
ICASSP '22	Sentiment-aware ASR Pre-training for Enhanced Speech Emotion Recognition Ghriss, A., Yang, B., Rozgic, V., Shriberg, E., Wang, C. SOTA on the valence (positivity) dimension at time of publication.
NeuroImage '22	Insights from an Autism Imaging Biomarker Challenge: Promises and Threats to Biomarker Discovery Traut, N., et al. (incl. Ghriss, A.) <i>Best ML model among participants.</i>
RIKEN AIP'18	Reinforcement Learning with Options and State Representation Ghriss, A., Sugiyama, M., Lazaric, A. <i>MVA Master's thesis.</i>

SELECTED OPEN-SOURCE SOFTWARE

PGCuts : Differentiable graph cuts with GPU-accelerated hypergeometric kernels.	pgcuts.readthedocs.io
SparseKit : Structured sparsity specification (S ³) toolkit, with Structured-OBS for LLM pruning.	sparsekit.readthedocs.io
Qwantize : Optimal quantization for block-scaled low-precision formats (NVFP4, MXFP4).	qwantize.readthedocs.io
Tritonix : Triton GPU kernels with an optimized autotuning framework.	ayghri/tritonix
Sphedron : Refinable polyhedral meshes on the sphere for GNN-based geospatial ML.	sphedron.readthedocs.io

WORK EXPERIENCE

Teaching Assistant	Aug 2021 – Present
CU Boulder, Computer Science - Graduate Machine Learning (CSCI 5266 / 5622): Fall '21, Fall '23, Spring '26. - Undergraduate Machine Learning (CSCI 4266), '22-'23; guest instructor for the Reinforcement Learning module. - Tutorials on Small Language Models (Fall '23). Outstanding Teaching Assistant Award, '22.	
Graduate Research Assistant	Jan 2024 – Dec 2025
CIRES, CU Boulder GNN-based emulators for climate-model dynamics on refinable spherical meshes; released as the <i>Sphedron</i> library.	
Visiting Researcher	Jun – Aug 2023
Inria, Paris Deep learning for optimization over discrete sets.	
Applied Scientist Intern	May – Aug 2022
Amazon Alexa Paralinguistics, Cambridge Speaker identification and paralinguistic speech-emotion modeling on conversational audio.	
Applied Scientist Intern	May – Aug 2021
Amazon Alexa Edge-ML, Cambridge Sentiment-aware ASR pre-training for on-device speech-emotion recognition (ICASSP '22).	
Software Engineer, PhD Intern	Jun – Aug 2020
Google, Boulder Deep learning for air-quality prediction from satellite imagery (Google Earth Engine).	
Research Intern	Apr – Sep 2018
RIKEN Advanced Intelligence Project, Tokyo Hierarchical deep reinforcement learning with options and learned state representations.	

SERVICE & TALKS

- **Reviewer:** ICLR 2025 • ICML 2025, 2026 • NeurIPS 2025
- **Tutorial Chair:** "Associative Memories", ICML 2025
- **Oral Presentation:** *Crowdsourced Information Authentication: A Graph-based Model from the Science of Hadith*, MusML Workshop, ICML 2025
- **Workshop Instructor:** *GPU Programming Workshop* – CS Graduate Professional Development Club, CU Boulder, Spring '26. Hands-on intro to parallel execution, thread/warp behavior, fused kernels, and Triton. github.com/ayghri/gpuwork
- **Guest Lectures:** *Introduction to Machine Learning* – ASEN 5307 Engineering Data Analysis Methods, CU Boulder (2 lectures in Fall '24, 2 in Fall '25)

AWARDS & HONORS

- **CU Boulder CS Dept. Outstanding Paper Award** – for *Beyond Spectral Clustering: Probabilistic Cuts for Differentiable Graph Partitioning* (AISTATS '26).
- **CU Boulder CS Dept. Outstanding Paper Award** – for *Deep Clustering via Probabilistic Ratio-Cut Optimization* (AISTATS '25).
- **Best Work-in-Progress Award**, CU Boulder CS Dept. Expo, Fall '25 – for work on structured sparse training (S³).
- **Outstanding Teaching Assistant Award**, CU Boulder CS Dept., '22.

EDUCATION

PhD, Machine Learning	2019 – 2026
🎓 University of Colorado, Boulder	
• Advisor: Claire Monteleoni. Expected graduation: Summer 2026.	
• Thesis: Structured Sparse Training and Differentiable Graph Cuts for Efficient ML.	
Research Master – Mathematics, Computer Vision, and Machine Learning (MVA)	Oct 2017 – Sep 2018
🎓 École Normale Supérieure Paris-Saclay	
• Reinforcement Learning, Probabilistic Graphical Models, Statistical Learning	
• Computer Vision, Deep Learning, Speech Recognition, Audio Signal Processing	
Diplôme d'Ingénieur – Polytechnicien cycle	Sep 2014 – Sep 2017
🎓 École Polytechnique	
• Polytechnician Engineer Cycle with Major in Applied and Computational Mathematics.	
• Stochastic Calculus, Monte Carlo methods, Advanced Machine Learning, Random Models of survival.	
B.Sc. Economics and Applied Mathematics	Sep 2013 – Jun 2014
🎓 ENSAE ParisTech	
• Data Analysis, Econometrics, Microeconomics, Macroeconomics	

LANGUAGES

Arabic	Native
French	Native
English	Fluent
Japanese	Intermediate
Chinese	Elementary

TECHNICAL SKILLS

GPU & Systems	CUDA	Triton
C/C++	Linux	Git
ML / DL	Python	PyTorch
	Scikit-learn	SciPy
Distributed	AWS	PySpark
	Horovod	MPI